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Human Kinesiology lab Report #1.

Name : Tom Wu (Yuxiang Wu).
Gender : Male.
Roll number: 23010P114.
Class : 23 Rehab
Date : 25/3/31
Title : Posture influence on cardiovascular parameters.

Background :

The CV system is responsible for maintaining adequate blood circulation. Various physiological and external factors influence CV parameters, including heart Rate, blood pressure and Cardiac output. Among these factors, body posture plays a significant role in CV regulation due to its impact on blood distribution and autonomic nervous system activity.

When a person transitions from a supine to an upright position, gravitational forces cause blood to pool in the lower extremities, leading to a temporary decrease in venous return and cardiac output. This drop in CV system, triggers compensatory mechanisms. Conversely, lying down reduces gravitational effects, facilitating venous return and increasing stroke volume.

This experiment aims to vestigate the effects of different body postures on CV parameters, analyzing factors and effects.

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object :

The object of this experiment is the effect of different body postures on cardiovascular parameters, specifically blood pressure and heart rate. The study focuses on two postures : Supine position and Supine position with elevated legs. By measuring BP and HR via upper arm and medial ankle and palpation, the experiment examines how postural changes influence hemodynamic regulation.

Method and Procedure :

1. Subjects :

This experiment involved 19 healthy adult subjects. All Subjects noticed to be rest least 5 minutes before measurements to ensure stable blood pressure and heart rate readings.

2. Equipment.

1. Mercury sphygmomanometer - for measuring blood pressure at the upper limb and lower limb.
2. Timer - measuring heart rate over one minute.
3. Record sheet - documenting the measurements on blackbroad

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3. Procedure

1. Meacurements in Supine Position.

① The subject lies in a supine position on a flat surface (bed or examination table), remaining relaxed for at least 2 minutes.

② BP is measured at the upper arm using mercury Sphygmomanometer, recording both systolic blood pressure and diastolic blood pressure.

③ Blood pressure is measured at the ankle, using the mercury sphygmanometer, recording SBP and DBP.

④ Heart rate is meacured by palpating the radial artery, recording the pulse rate over one minute.

2. Measurements in Spine Position with Elevated Legs.

①. subject remain Supine position, but both legs are elevated to about 45 degrees, maintaining this posture for at least 2minutes to allow experiment processing. Rest steps Repeat the measurments in Supine Position.

3. At the end of the experiment, the height and weight of each subject were recorded, and corresponding BMI was calculated.

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4. Data Processing // Statistics.

①. Contains two worksheets.

23 Rehab origin data : Raw data from 18 subjects, including demographics (age, gender, hight, weight, BMI) and blood pressure measurements (Systolic, diastolic, heart rate, pulse pressure) in two positions (lying supine, lower limb lift), recorded at Arm and Ankle.

T test : Paired t-tests comparing BP paraments between Arm and Ankle measurements.

Data shows in Raw excle.

5 Results

1、key observations.

① Ankle systolic BP Decreases with Limb Lift.

Significant drop with 8 mmHg ($P = 0.0115$) when legs were Raised! Suggests altered peripheral resistance or Venous return.

② No Significant Changes in Diastolic BP or pulse Pressure. Diastolic BP remained stable ($p = 0.6096$), Pulse showed no significant Variation ($p = 0.2854$).

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③ Gender differences

Males had higher systolic BP (124.67 vs. 110.08 mmHg in females). Potential BMI influence (male 23.21 vs. female 21.14) with total Mean BMI 21.82±0.83.

6. Discussion

1. Hemodynamic Implications.

Ankle BP drop During Limb lift, likely due to reduced venous pooling and increased central blood volume. Alingns with known postural hemodynamic responses.

2. Stable pulse pressure.

Indicates maintained cardiac output and arterial Compliance.

3. Gender differences.

Higher BP in Males, May reflect higher BMI or physiological sex differences. Limited male Sample (n=6) requires cautious interpretation.

4. Limitations.

Small sample size, especially for males, reducing Statistical power. With missing data, they might possibly affecting affecting analysis. And Activity level, hydration, and time of measurement are not Standardized. Some objects did not show significant data for BP, recorded by observed of floating of Meniscus of mercury column.

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7. Conclusion.

Limb lifting significantly reduces ankle systolis BP, likely due to hemodynamic redistribution. Diastolic BP and pulse pressure remain stable, suggesting preserved vascular function. Males exhibit high BP, possibly linked to BMI.

In future, Larger, gender-balanced studies are needed. Include continuous BP monitoring during posture changes. Advised investigate long-term effects of repeated limb lifting on circulation.

8. References:

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(2) Connelly, Paul J et al. "Transgender adults, gender-affirming hormone therapy and blood pressure: a systematic review." Journal of hypertension Vol. 39, 2 (2021): 223-230. doi: 10.1097/HJH.0000000000002632

(3) Lin, Hairong et al. "Arm Position and Blood pressure Readings." JAMA internal medicine Vol. 184, 12 (2024): 1436-1442. doi: 10.1001/jamainternmed.2024.5213